

Question 1

T5

Gina correctly started to answer the following question. Complete her solution.

Question: Solve the following equation for all real values of θ .
Express your answer in radians correct to 3 decimal places.

$$3\sin^2 \theta - 14\sin \theta - 5 = 0$$

Gina's solution: $3\sin^2 \theta - 14\sin \theta - 5 = 0$
 $(3\sin \theta + 1)(\sin \theta - 5) = 0$

Solution

Method 1

$$3\sin^2 \theta - 14\sin \theta - 5 = 0$$

$$(3\sin \theta + 1)(\sin \theta - 5) = 0$$

$$\sin \theta = -\frac{1}{3} \qquad \sin \theta = 5$$

no solution

$\frac{1}{2}$ mark for $\sin \theta = 5$
 $\frac{1}{2}$ mark for no solution

$$\theta_r = 0.339\ 837$$

$$\theta = 3.481\ 429, 5.943\ 348$$

1 mark ($\frac{1}{2}$ mark for each value of θ)

$$\theta = 3.481 + 2k\pi, k \in \mathbb{I}$$

$$\theta = 5.943 + 2k\pi, k \in \mathbb{I}$$

1 mark for general solution

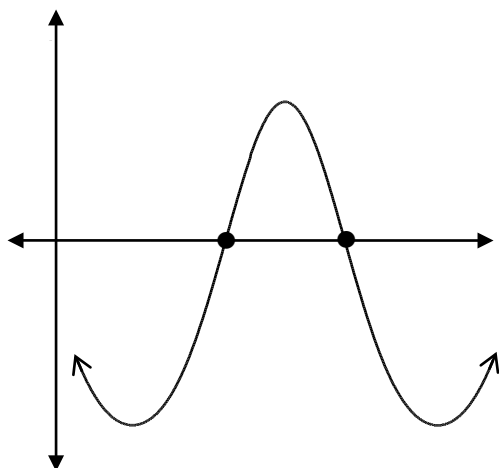
3 marks

Note(s):

- give maximum of 2 marks for solution in degrees: $\theta = 199.471^\circ + 360^\circ k, k \in \mathbb{I}$
 $\theta = 340.529^\circ + 360^\circ k, k \in \mathbb{I}$

Method 2

$$y = 3 \sin^2 \theta - 14 \sin \theta - 5$$

 $\frac{1}{2}$ mark for equation**or** $\frac{1}{2}$ mark for justification

Find all zeros over the reals.

$$\theta = 3.481 + 2k\pi, k \in \mathbb{I}$$

1 mark for solutions

$$\theta = 5.943 + 2k\pi, k \in \mathbb{I}$$

1 mark for general solution

3 marks

Find and simplify the 6th term in the binomial expansion of $\left(3x^4 - \frac{1}{x^3}\right)^9$.

Solution

$$\begin{aligned}t_6 &= {}_9C_5 \left(3x^4\right)^4 \left(-\frac{1}{x^3}\right)^5 \\&= (126)(81x^{16})\left(-\frac{1}{x^{15}}\right) \\&= -10\,206x\end{aligned}$$

2 marks (1 mark for ${}_9C_5$, $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for evaluating coefficient, $\frac{1}{2}$ mark for simplifying variable)

3 marks

Question 3

R10

The number of times a website is visited can be modeled by the function:

$$A = 800(e)^{rt}$$

where A = the total number of visitors at time t

t = the time in days ($t \geq 0$)

r = the rate of growth

After 5 days, 40 000 people have visited the site.

Determine the number of visitors expected after 9 days.

Express your answer as a whole number.

Solution

Method 1

$$40\,000 = 800e^{r5}$$

½ mark for substitution

$$\frac{40\,000}{800} = e^{5r}$$

$$\ln 50 = \ln e^{5r}$$

½ mark for applying logs

$$\ln 50 = 5r$$

1 mark for log theorem

$$\frac{\ln 50}{5} = r$$

$$r = 0.782\,404\,601$$

$$A = 800e^{(0.782\,404\,601)(9)}$$

½ mark for substitution

$$A = 914\,610.103$$

½ mark for calculations with base e

$$A = 914\,610$$

3 marks

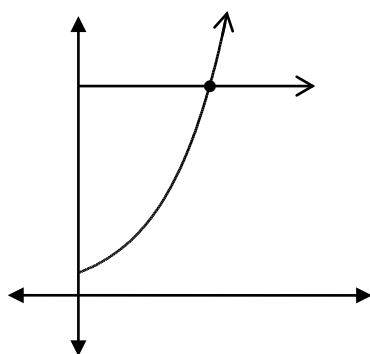
Method 2

Use calculator to find the value of r .

$$y = 40\,000$$

$\frac{1}{2}$ mark for equations

$$y = 800e^{5x}$$



or

$\frac{1}{2}$ mark for justification

Find the point of intersection of these two functions.

$$x = 0.782\,404\,601$$

1 mark for finding the value of x at the intersection point

Substitute x for r into formula.

$$\therefore A = 800e^{(0.782\,404\,601)(9)}$$

$$A = 914\,610.103$$

$$A = 914\,610$$

1 mark for value of A

3 marks

Question 4

R10

Solve algebraically:

$$10^{3x} = 7^{x+5}$$

Express your answer correct to 3 decimal places.

Solution

Method 1

$$10^{3x} = 7^{x+5}$$

$$\log 10^{3x} = \log 7^{x+5}$$

$$3x(\log 10) = (x+5)\log 7$$

$$3x \log 10 = x \log 7 + 5 \log 7$$

$$3x \log 10 - x \log 7 = 5 \log 7$$

$$x = \frac{5 \log 7}{3 \log 10 - \log 7}$$

$$x = 1.960\ 873$$

$$x = 1.961$$

½ mark for applying logs

1 mark for power rule

½ mark for collecting terms with x

½ mark for solving for x

½ mark for evaluating quotient of logarithms

Method 2

$$10^{3x} = 7^{x+5}$$

$$\log 10^{3x} = \log 7^{x+5}$$

$$3x(\log 10) = (x+5)\log 7$$

$$3x = x \log 7 + 5 \log 7$$

$$3x - x \log 7 = 5 \log 7$$

$$x = \frac{5 \log 7}{3 - \log 7}$$

$$x = 1.960\ 873$$

$$x = 1.961$$

3 marks

Question 5

P2

A word contains two Ms, two Es, two Ns, and no other repeated letters.

Suppose one of the Ns is replaced by an M.

Will this replacement result in greater or fewer permutations?

Justify your reasoning.

Solution

Method 1

$$2 \text{ Ms, } 2 \text{ Es, } 2 \text{ Ns: } \frac{(\text{total number of letters})!}{2!2!2!}$$

1 mark for dividing the total number of letters by $2!2!2!$

$$\frac{(\text{total number of letters})!}{8}$$

$$3 \text{ Ms, } 2 \text{ Es, } 1 \text{ N: } \frac{(\text{total number of letters})!}{3!2!1!}$$

1 mark for dividing the total number of letters by $3!2!1!$

$$\frac{(\text{total number of letters})!}{12}$$

2 marks

If one of the Ns is changed to an M, there would be fewer permutations.

Method 2

If one of the Ns is changed to an M, there would be fewer permutations because you would be dividing the total number of letters by a larger number.

2 marks for justification

2 marks

Question 6

P3

There is a group of 16 boys and 12 girls. How many ways can a committee of 3 people be formed if there must be at least 2 girls on the committee?

Express your answer as a whole number.

Solution

Method 1

Case 1: 2 girls, 1 boy

$${}_{12}C_2 \cdot {}_{16}C_1 = 1056$$

1 mark for case 1 ($\frac{1}{2}$ mark for each factor shown in a product)

Case 2: 3 girls

$${}_{12}C_3 = 220$$

1 mark for case 2

$$1056 + 220 = 1276$$

1 mark for addition of cases

3 marks

Method 2

Case 1: 2 boys, 1 girl

$${}_{16}C_2 \cdot {}_{12}C_1 = 1440$$

1 mark for case 1 ($\frac{1}{2}$ mark for each factor shown in a product)

Case 2: 3 boys

$${}_{16}C_3 = 560$$

1 mark for case 2

All possibilities

$${}_{28}C_3 = 3276$$

$\frac{1}{2}$ mark for all possible cases

$$3276 - 1440 - 560 = 1276$$

$\frac{1}{2}$ mark for subtraction from total

3 marks

Question 7

T1

A student is using the formula $s = \theta r$ to find an arc length of a circle. Given a central angle measure of 35° and a radius of 6 cm, the student's solution is as follows:

$$s = (35)(6)$$

$$s = 210 \text{ cm}$$

Explain why this solution is incorrect.

Write the correct solution.

Solution

When using the formula $s = \theta r$, the angle must be in radians. $\frac{1}{2}$ mark for explanation

$$35^\circ \left(\frac{\pi}{180^\circ} \right) = \frac{35\pi}{180} \quad \text{or} \quad \frac{7\pi}{36}$$

1 mark for conversion to radians

$$\therefore s = \theta r$$

$$= \frac{7\pi}{36}(6)$$

$$= \frac{42\pi}{36} \text{ cm} \quad \text{or} \quad \frac{7\pi}{6} \text{ cm} \quad \text{or} \quad 3.665 \text{ cm}$$

$\frac{1}{2}$ mark for simplification

2 marks

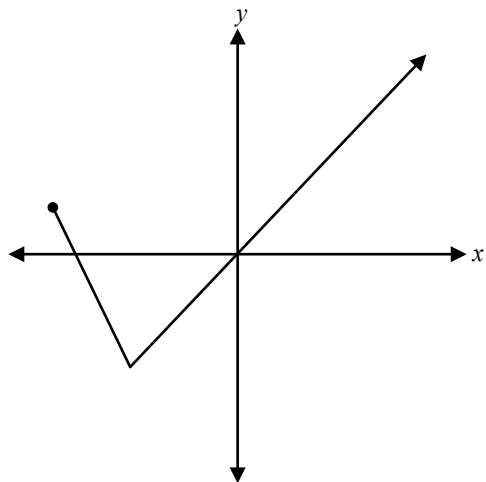
Note(s):

- award 1 mark for correct final answer with no work

Question 8

R1

Given the graph of $f(x)$ below, explain how you would sketch the graph of $y = |f(x)|$.



Solution

The negative y values are reflected over the x -axis.

1 mark for explanation

1 mark

Question 9

R10

Claire correctly solves the following equation:

$$\log_2(6-x) + \log_2(3-x) = 2$$

She finds two possible values of x : $x = 2$ and $x = 7$.

Identify which one of these values is unacceptable and explain why.

Solution

If x is greater than 3, you have a negative argument $\therefore x = 2$ but $x \neq 7$.

or

1 mark for explanation

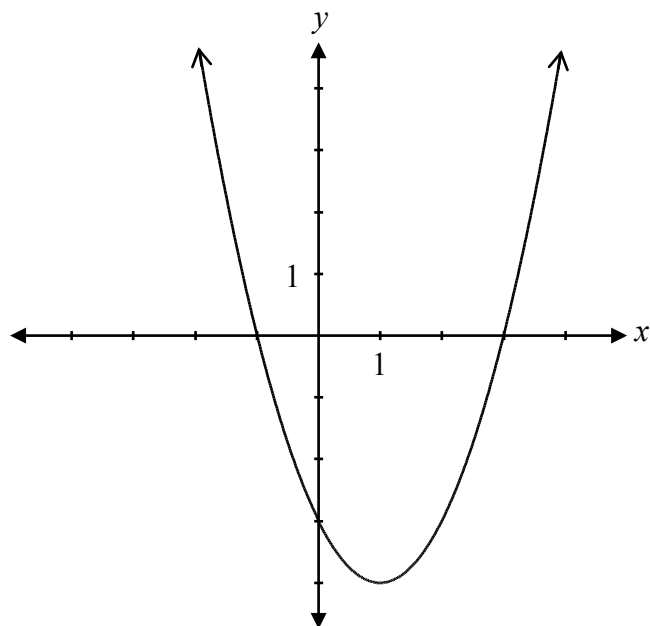
The domain is restricted to values of $x < 3 \therefore x = 2$.

1 mark

Question 10

R13

Given the graph of the function $f(x)$ below, state the domain of $y = \sqrt{f(x)}$.



Solution

The domain is $\{x \in \mathbb{R} \mid x \leq -1 \text{ or } x \geq 3\}$.

or

The domain is $(-\infty, -1] \cup [3, \infty)$.

1 mark for domain

1 mark

Question 11

P1

A school offers 4 different Science courses, 3 different Mathematics courses, and 2 different English courses.

Julie must select 1 Science course, 1 Mathematics course, and 1 English course. She thinks this creates 9 options for her timetable.

Show why Julie is incorrect.

Solution

Julie should have multiplied the number of options.

1 mark for justification

or

1 mark

$4 \times 3 \times 2 = 24$ different options

Explain how Pascal's triangle can be used to determine the coefficients in the binomial expansion of $(x + y)^n$.

Solution

The $(n + 1)^{\text{st}}$ row of Pascal's triangle corresponds to the coefficients of the terms in the expansion of the binomial $(x + y)^n$.

1 mark for explanation

1 mark

Prove the identity below for all permissible values of x :

$$\frac{1 + \cos 2x}{\sin 2x} = \cot x$$

Solution

Method 1

$$\begin{aligned} \text{LHS} &= \frac{1 + 2\cos^2 x - 1}{2\sin x \cos x} && \begin{array}{l} 1 \text{ mark for identity} \\ \frac{1}{2} \text{ mark for identity} \end{array} \\ &= \frac{2\cos^2 x}{2\sin x \cos x} \\ &= \frac{\cos x}{\sin x} && 1 \text{ mark for simplification} \\ &= \cot x && \frac{1}{2} \text{ mark for identity} \\ &= \text{RHS} \end{aligned}$$

3 marks

Method 2

$$\begin{aligned} \text{LHS} &= \frac{1 + 1 - 2\sin^2 x}{2\sin x \cos x} && \begin{array}{l} \frac{1}{2} \text{ mark for identity} \\ \frac{1}{2} \text{ mark for identity} \end{array} \\ &= \frac{2 - 2\sin^2 x}{2\sin x \cos x} \\ &= \frac{1 - \sin^2 x}{\sin x \cos x} && \frac{1}{2} \text{ mark for simplification} \\ &= \frac{\cos^2 x}{\sin x \cos x} && \frac{1}{2} \text{ mark for identity} \\ &= \frac{\cos x}{\sin x} && \frac{1}{2} \text{ mark for simplification} \\ &= \cot x && \frac{1}{2} \text{ mark for identity} \\ &= \text{RHS} \end{aligned}$$

3 marks

Solution**Method 3**

$$\text{LHS} = \frac{1 + \cos^2 x - \sin^2 x}{2 \sin x \cos x}$$

½ mark for identity

½ mark for identity

$$= \frac{2 \cos^2 x}{2 \sin x \cos x}$$

½ mark for identity $(1 - \sin^2 x = \cos^2 x)$

$$= \frac{\cos x}{\sin x}$$

1 mark for simplification

$$= \cot x$$

½ mark for identity

$$= \text{RHS}$$

3 marks

Your classmate, Leo, was absent for one of his math lessons.

Explain to Leo how to determine the cosecant ratio for an angle in standard position given that $P(-3, -4)$ is a point on the terminal arm of the angle.

Solution

Method 1

- Locate the point $(-3, -4)$ on the coordinate plane.
- Draw a right-angle triangle by connecting the point $(-3, -4)$ to the x -axis and to the origin.
- The angle created with the x -axis and the line connecting the point with the origin (the terminal arm) is θ . ½ mark
- Determine the length of the terminal arm by using the Pythagorean Theorem. ½ mark
- Once you have the lengths of all three sides of the triangle, find $\sin \theta$. ½ mark
- Invert the $\sin \theta$ ratio to find $\csc \theta$. ½ mark

2 marks

Method 2

- Determine the value of r using the Pythagorean Theorem. ½ mark
- Determine the value of $\sin \theta \left(\sin \theta = \frac{y}{r} \right)$. ½ mark
- Identify the correct quadrant. ½ mark
- Invert the $\sin \theta$ ratio to find $\csc \theta$. ½ mark

2 marks

Note(s):

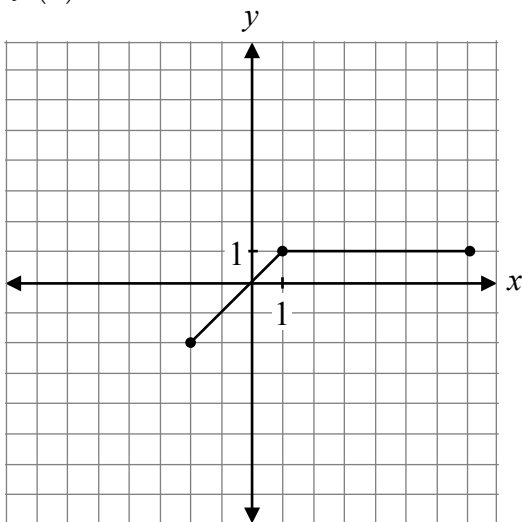
- award a maximum of 1 mark for correct work shown without an explanation in words

Question 15

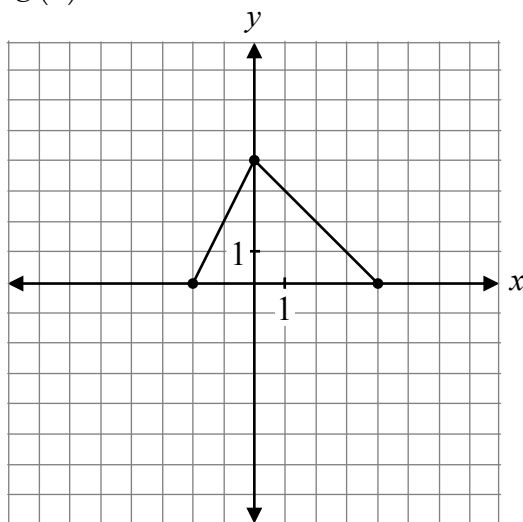
R1

Given the following graphs:

$f(x)$

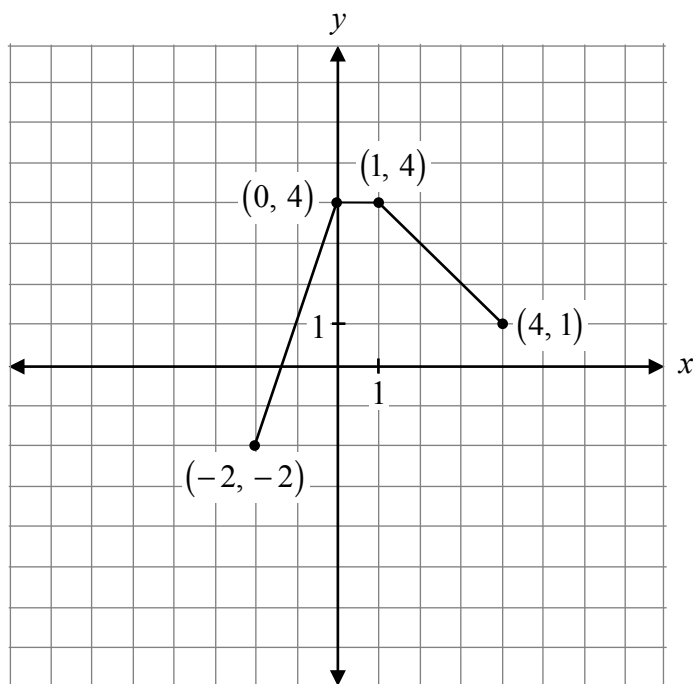


$g(x)$



Sketch the graph of $f(x) + g(x)$.

Solution



2 marks ($\frac{1}{2}$ mark for each point where the graph changes direction)

$[(-2, -2); (0, 4); (1, 4); (4, 1)]$

2 marks

Note(s):

- deduct a maximum of 1 mark for incorrect domain

Question 16

R3

If $(2, 3)$ is a point on the graph of $y = f(x)$, what point must be on the graph of $y = 3f\left(\frac{1}{4}x\right)$?

a) $\left(\frac{1}{2}, 1\right)$

b) $\left(\frac{1}{2}, 9\right)$

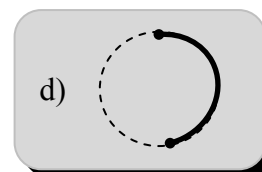
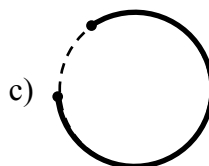
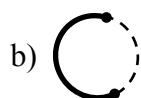
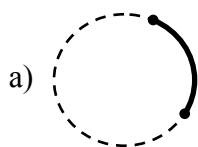
c) $(8, 1)$

d) $(8, 9)$

Question 17

T1

Consider the arc drawn on each circle. Which arc measure is closest to 3 radians?



Question 18

R7

If $\log_2 x = 4$, then $\log_2 (2x)$ is equal to:

a) 5

b) 8

c) 16

d) 32

Question 19

T6

Simplify the following expression:

$$\cos^2 x (1 + \cot^2 x)$$

a) $\sin^2 x$

b) $\cos^2 x$

c) $\cot^2 x$

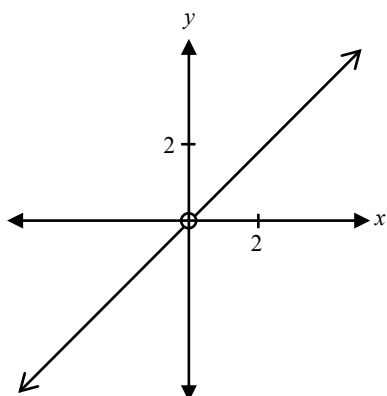
d) $\sec^2 x$

Question 20

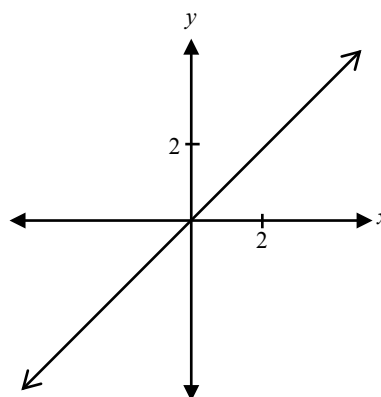
R14

Identify the graph of the function $y = \frac{x}{x}$.

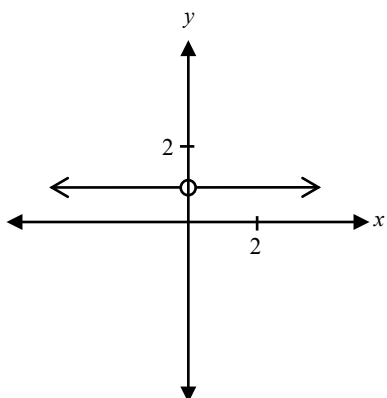
a)



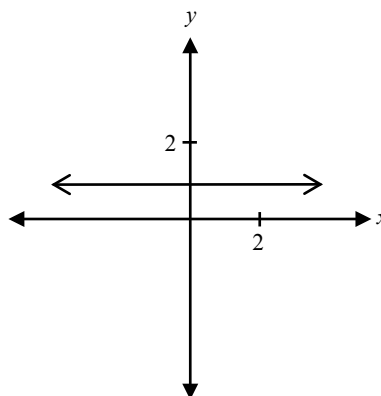
b)



c)



d)



Question 21

P4

How many terms are in the expansion of $(3y^2 - 4z)^7$?

a) 2

b) 6

c) 7

d) 8

Question 22

R6

Determine one possible restriction for the domain of $y = (x + 3)^2 - 4$ so that its inverse is a function.

a) $x \leq -3$

b) $x \leq 0$

c) $x \leq 3$

d) $x \leq 4$

Question 23

P2

Find the total possible number of arrangements for 7 adults and 3 children seated in a row if the 3 children must sit together.

a) $10!$

b) $8!3!$

c) $7!3!$

d) $7!$

Question 24

R9

Identify the value of the x -intercept of the function $y = \ln(x - 2)$.

a) -1

b) 0

c) 2

d) 3

Question 25

R7

Given $\log_b a = 3$, give one example of possible values for a and b that make this equation true.

Solution

Answers will vary but $b^3 = a$.

Some possible solutions are: $a = 8$ $b = 2$

or

$a = 27$ $b = 3$

or

$a = 64$ $b = 4$

1 mark

Question 26

R5

The range of the graph of $y = f(x)$ is $[-3, 2]$.

Explain why there is no effect on the range of the graph that is a result of the transformation $y = f(-x)$.

Solution

$y = f(-x)$ is a reflection over the y -axis.

The domain is affected, but the range remains the same. 1 mark for explanation

1 mark

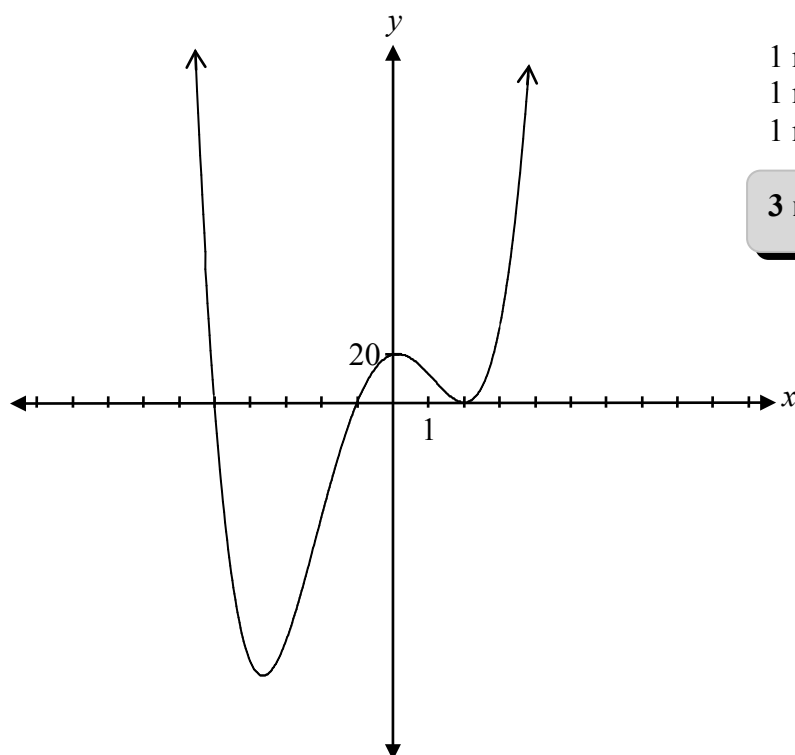
Sketch the graph of $y = (x + 1)(x - 2)^2(x + 5)$.

Identify the x -intercepts and y -intercept.

Solution

x -intercepts: -5 , -1 , and 2

y -intercept: 20



1 mark for x -intercepts

1 mark for y -intercept

1 mark for multiplicity of a zero at $x = 2$

3 marks

Note(s):

- if no graph is shown, give $\frac{1}{2}$ mark for x -intercepts (-5 , -1 , and 2) and $\frac{1}{2}$ mark for y -intercept (20)
- relative maximum and minimum are not required

Question 28

T4

The graph of the function $y = \sin x$ has been transformed to create a new graph.

The range of this new graph is $[-4, 4]$ and the zeros are $x = k\frac{\pi}{2}$, where k is an integer.

Write the equation that corresponds to this new graph.

Solution

Amplitude = 4

1 mark for correct amplitude

Period = π

$\frac{1}{2}$ mark for correct period

$$\therefore b = \frac{2\pi}{\pi} = 2$$

$\frac{1}{2}$ mark for consistent value of b

2 marks

$$y = 4 \sin(2x)$$

or

$$y = -4 \sin(2x)$$

Note(s):

- deduct $\frac{1}{2}$ mark for a translation that results in an incorrect range and/or zeros

Question 29

R1

Given the functions $f(x) = x^2 - 1$ and $g(x) = x + 1$, state the domain of $\frac{g(x)}{f(x)}$.

Solution

Domain: $x \in \mathbb{R}$ where $x \neq 1$ and $x \neq -1$

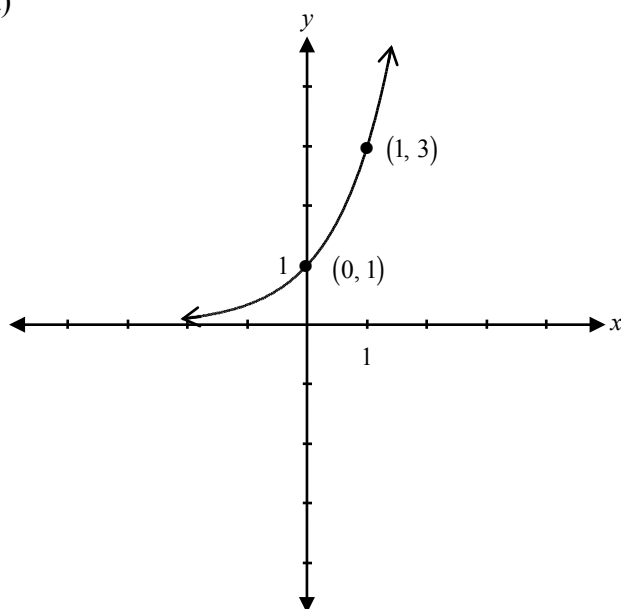
1 mark ($\frac{1}{2}$ mark for $x \neq 1$, $\frac{1}{2}$ mark for $x \neq -1$)

1 mark

- a) Sketch the graph of $y = 3^x$.
- b) Explain how the graph of $y = 3^x$ can be used to sketch the graph of $y = \log_3 x$.

Solutions

a)



½ mark for increasing exponential function
½ mark for y-intercept at (0, 1)
½ mark for consistent point on exponential function
½ mark for asymptotic behaviour

2 marks

b)

To graph $y = \log_3 x$, you can reflect the graph of $y = 3^x$ over the line $y = x$.

or

You can switch the x and y coordinates of $y = 3^x$ to get the graph of $y = \log_3 x$.

1 mark for explanation

1 mark

Note(s):

- in (b), give ½ mark for having only stated they are inverse functions of each other

Question 31

R11, R12

A box in the shape of a rectangular prism has side lengths x , $x + 2$, and $x + 10$.

Write a function, $V(x)$, to express the volume of the box in terms of x .

Find all possible values of x , given that the volume of the box is 96 cm^3 .

State the dimensions of the box.

Solution

$$V(x) = (x)(x+2)(x+10)$$

$\frac{1}{2}$ mark for expressing volume in terms of x

$$96 = (x)(x+2)(x+10)$$

$$96 = x^3 + 12x^2 + 20x$$

$$0 = x^3 + 12x^2 + 20x - 96$$

$\frac{1}{2}$ mark for equating to zero

when $x = 2$:

$$0 = (2)^3 + 12(2)^2 + 20(2) - 96$$

$$0 = 8 + 48 + 40 - 96$$

$$0 = 0$$

$\therefore x = 2$ is a possible value.

1 mark for identifying one possible value of x

$$\begin{array}{r|rrrr} 2 & 1 & 12 & 20 & -96 \\ & & 2 & 28 & 96 \\ \hline & 1 & 14 & 48 & 0 \end{array}$$

1 mark for division

$$(x-2)(x^2 + 14x + 48) = 0$$

$$(x-2)(x+8)(x+6) = 0$$

$$x = -8, -6, 2$$

1 mark for identifying all possible values of x

$x \neq -8$ and $x \neq -6$ because dimensions cannot be negative values.

$\therefore x = 2$ is the only solution.

$\frac{1}{2}$ mark for rejecting extraneous roots

The dimensions of the box are $2 \text{ cm} \times 4 \text{ cm} \times 12 \text{ cm}$.

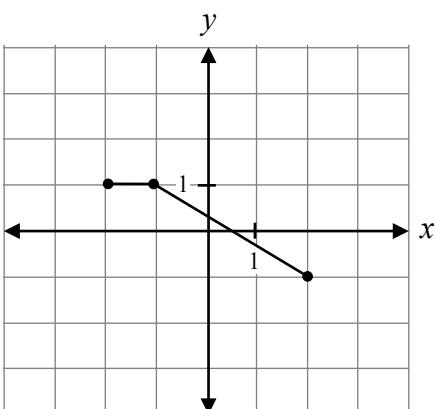
$\frac{1}{2}$ mark for stating the dimensions of the box

5 marks

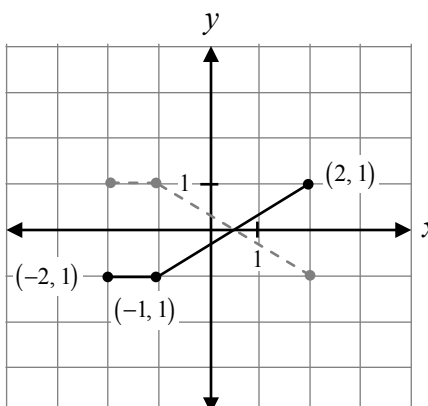
Question 32

R5

Given the graph of $f(x)$ below, sketch the graph of $y = -f(x)$.



Solution



1 mark

Question 33

T2

Determine the coordinates of a point (x, y) on the unit circle if you are given $\theta = 30^\circ$ where θ is in standard position.

Solution

$$P(\theta) = (\cos \theta, \sin \theta)$$

If $\theta = 30^\circ$, then the coordinates of $P(\theta)$ would be $(\cos 30^\circ, \sin 30^\circ)$.

or

$$P(30^\circ) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$$

1 mark for answer stated
as coordinate point

1 mark

Given the following sinusoidal equation:

$$P(t) = 3000 \sin\left[\frac{\pi}{10}(t - 2010)\right] + 10\,000$$

Determine the maximum value of $P(t)$ and a value of t at which this maximum occurs.

Solution

Method 1

$$\begin{aligned}\text{Maximum value} &= 10\,000 + 3000 \\ &= 13\,000\end{aligned}$$

1 mark for maximum value

$$13\,000 = 3000 \sin\left[\frac{\pi}{10}(t - 2010)\right] + 10\,000$$

$$3000 = 3000 \sin\left[\frac{\pi}{10}(t - 2010)\right]$$

$$1 = \sin\left[\frac{\pi}{10}(t - 2010)\right]$$

$\frac{1}{2}$ mark for simplifying

$$\frac{\pi}{2} = \frac{\pi}{10}(t - 2010)$$

1 mark for exact value

$$5 = t - 2010$$

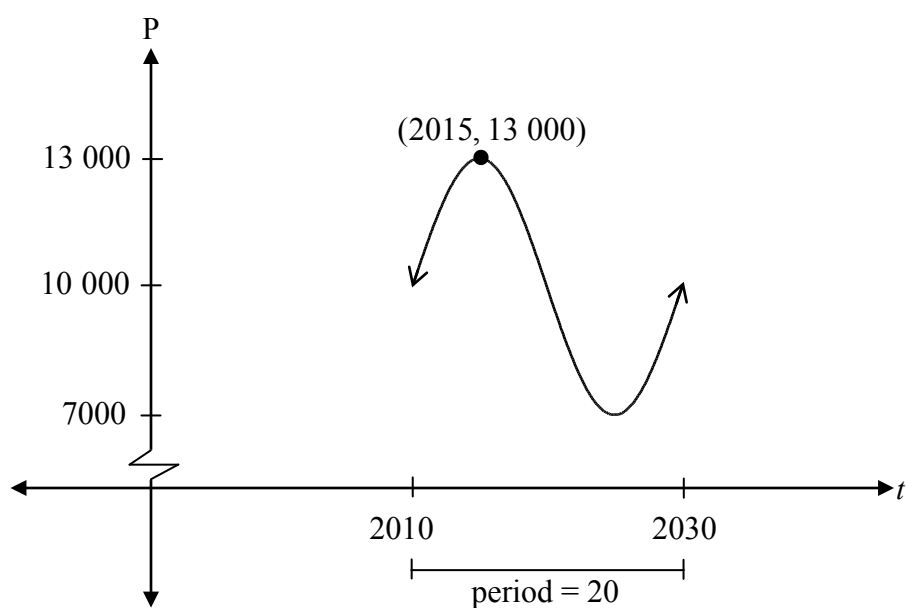
$$t = 2015$$

$\frac{1}{2}$ mark for solving for t

3 marks

Note(s):

- the period of the function is 20 $\therefore$ other acceptable answers are: $t = 2015 \pm 20$

Solution**Method 2**

1 mark for period

$$P(t) = 13\ 000$$

1 mark for maximum value

$$t = 2015$$

1 mark for solving for t

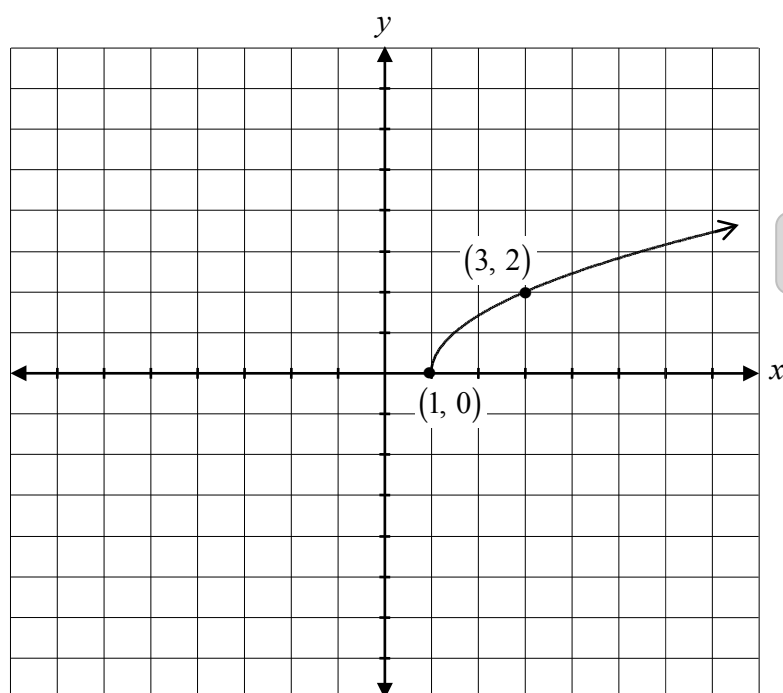
$\therefore$ the maximum value is 13 000 when $t = 2015$.

3 marks

Sketch the graph of $y = \sqrt{2x - 2}$.

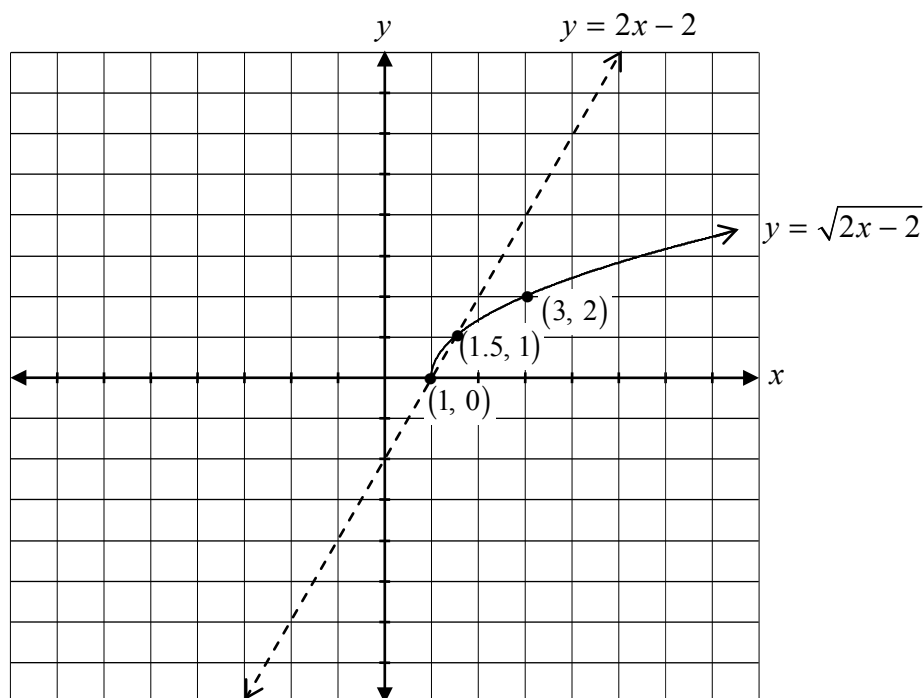
Solution**Method 1**

$$\begin{aligned} y &= \sqrt{2x - 2} \\ &= \sqrt{2(x - 1)} \end{aligned}$$



1 mark for domain: $[1, \infty)$
1 mark for shape (graph of a radical function)
1 mark for horizontal compression

3 marks

Method 2

1 mark for domain of $y = \sqrt{2x - 2} : [1, \infty)$

1 mark for invariant points where $y = 0$ and $y = 1$ ($\frac{1}{2}$ mark for each point)

$\frac{1}{2}$ mark for graph of $y = \sqrt{2x - 2}$ drawn above the graph of $y = 2x - 2$ between the invariant points

$\frac{1}{2}$ mark for graph of $y = \sqrt{2x - 2}$ drawn below the graph of $y = 2x - 2$ after the invariant point where $y = 1$

3 marks

Question 36

R6

Given $f(x) = 2x - 6$, write the equation of $f^{-1}(x)$.

Solution

Switch the x and y values.

$$x = 2y - 6$$

1 mark for switching x and y values

$$x + 6 = 2y$$

$$\frac{x + 6}{2} = y$$

$\frac{1}{2}$ mark for solving for y

$$\therefore f^{-1}(x) = \frac{x + 6}{2}$$

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

2 marks

Question 37

R8

Frank tried to expand a logarithmic expression using the laws of logarithms. He made one error.

$$\text{Frank's solution: } \log_a \frac{(x + 2)}{zw} = \log_a x + \log_a 2 - \log_a z - \log_a w$$

Write the correct solution.

Solution

$$\text{Correct solution: } \log_a \frac{(x + 2)}{zw} = \log_a (x + 2) - \log_a z - \log_a w$$

1 mark for correct solution

1 mark

Question 38

T6

Determine all non-permissible values of θ over the interval $[0, 2\pi]$.

$$\frac{\sin \theta}{1 + \cos \theta} + \csc \theta + \cot \theta$$

Explain your reasoning.

Solution

To determine non-permissible values, the denominator needs to equal zero.

1 mark for explanation

The denominators of this expression are “ $1 + \cos \theta$ ” and “ $\sin \theta$ ” $\left(\text{since } \csc \theta = \frac{1}{\sin \theta} \text{ and } \cot \theta = \frac{\cos \theta}{\sin \theta} \right)$.

1 mark ($\frac{1}{2}$ mark for identifying each restriction)

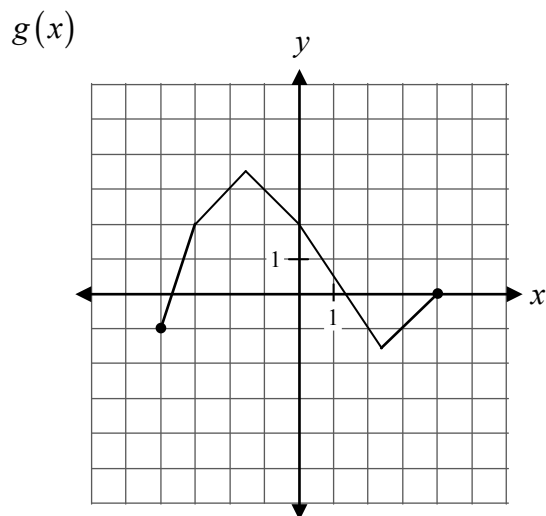
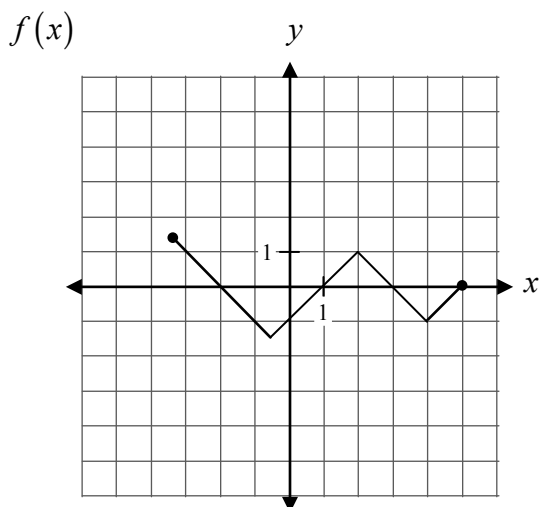
$$\begin{aligned} \therefore 1 + \cos \theta &= 0 & \sin \theta &= 0 \\ \cos \theta &= -1 & \theta &= 0, \pi, 2\pi \\ \theta &= \pi \end{aligned}$$

1 mark for all non-permissible values of θ ($\frac{1}{2}$ mark for each equation)

$\therefore$ the non-permissible values of θ over $[0, 2\pi]$ are $0, \pi$, and 2π .

3 marks

Given the following graphs:



- Determine the value of $[f \cdot g](0)$.
- Determine the value of $g(f(4))$.
- Determine a value for k where $f(k)=1$.

Solutions

$$\begin{aligned} \text{a) } f(0) &= -1 \\ g(0) &= 2 \end{aligned}$$

$$\begin{aligned} [f \cdot g](0) &= (-1)(2) \\ &= -2 \end{aligned}$$

1 mark for the value of $[f \cdot g](0)$

1 mark

$$\begin{aligned} \text{b) } f(4) &= -1 \\ g(-1) &= 3 \end{aligned}$$

$\frac{1}{2}$ mark for $f(4)$

$\frac{1}{2}$ mark for $g(f(4))$ consistent with $f(4)$ value

1 mark

$$\text{c) } k = 2 \quad \text{or} \quad k = -3$$

1 mark for a value for k

1 mark

Question 40

R1

Given that $h(x) = 2x^2 + 5x - 3$ and that $h(x) = f(x) \cdot g(x)$, determine $f(x)$ and $g(x)$.

Solution

$$f(x) = 2x - 1$$

$$g(x) = x + 3$$

1 mark for two correct factors of $h(x)$

1 mark

Other answers are possible.

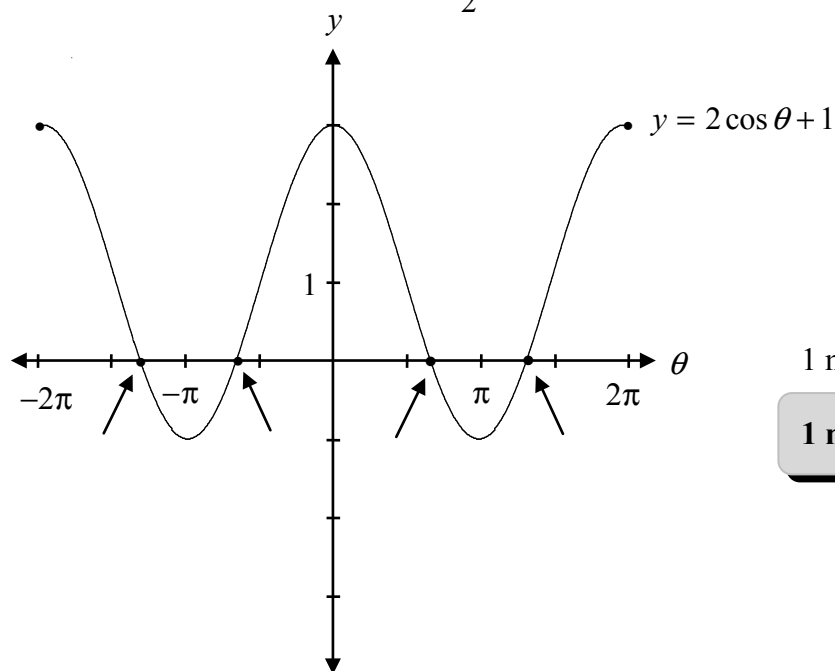
Question 41

T5

The graph of $y = 2 \cos \theta + 1$ below can be used to solve the equation $\cos \theta = -\frac{1}{2}$ over the interval $[-2\pi, 2\pi]$. Indicate on the graph where to find the solutions to the equation $\cos \theta = -\frac{1}{2}$.

Solution

The solution to the equation $\cos \theta = -\frac{1}{2}$ is found where the graph of $y = 2 \cos \theta + 1$ crosses the x-axis.



1 mark for indicating solutions on graph

1 mark

Note(s):

- give $\frac{1}{2}$ mark for indicating 1, 2, or 3 of the solutions

Question 42

R1

The function $f(x)$ is transformed.

A new function, $y = \frac{1}{f(x)}$, is created that does not have any vertical asymptotes.

What can you conclude about the original function $f(x)$?

Solution

If $f(x)$ does not have any x -intercepts, then the transformation would not have any vertical asymptotes.

or

$f(x)$ cannot equal zero.

1 mark for conclusion

1 mark

or

$f \neq 0$

Note(s):

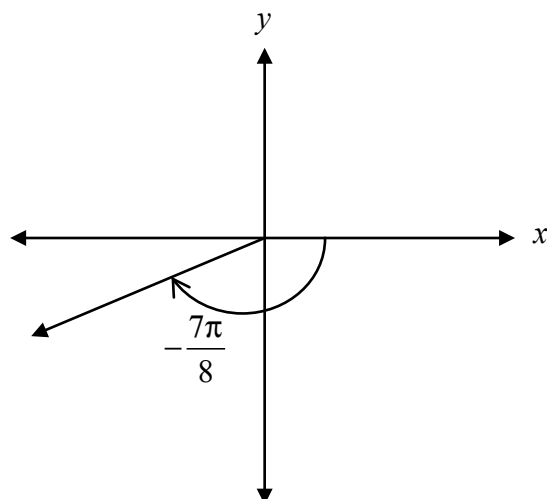
- award $\frac{1}{2}$ mark for a correct example without a given conclusion

Question 43

T1

Draw the angle $-\frac{7\pi}{8}$ in standard position.

Solution



1 mark for angle drawn in Quadrant III

1 mark

Determine the exact value of:

$$4 \cos\left(\frac{11\pi}{12}\right)$$

Solution

Method 1

$$\begin{aligned} 4 \cos\left(\frac{11\pi}{12}\right) &= 4 \cos\left(\frac{\pi}{4} + \frac{2\pi}{3}\right) \\ &= 4 \left[\cos \frac{\pi}{4} \cos \frac{2\pi}{3} - \sin \frac{\pi}{4} \sin \frac{2\pi}{3} \right] \\ &= 4 \left[\left(\frac{\sqrt{2}}{2}\right) \left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) \right] \\ &= 4 \left[\frac{-\sqrt{2} - \sqrt{6}}{4} \right] \\ &= -\sqrt{2} - \sqrt{6} \end{aligned}$$

1 mark for combination

2 marks ($\frac{1}{2}$ mark for each exact value)

3 marks

Solution**Method 2**

$\frac{11\pi}{12}$ has a reference angle of $\frac{\pi}{12}$.

½ mark for correct reference angle

$$\begin{aligned}\cos\left(\frac{\pi}{12}\right) &= \cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right) \\ &= \cos\frac{\pi}{4}\cos\frac{\pi}{6} + \sin\frac{\pi}{4}\sin\frac{\pi}{6} \\ &= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) \\ &= \frac{\sqrt{6} + \sqrt{2}}{4}\end{aligned}$$

½ mark for substitution into correct identity

1 mark for exact values

Since $\frac{11\pi}{12}$ is in Quadrant II, cosine is negative.

$$\begin{aligned}\therefore 4\cos\left(\frac{11\pi}{12}\right) &= 4\left(\frac{-\sqrt{6} - \sqrt{2}}{4}\right) \\ &= -\sqrt{6} - \sqrt{2}\end{aligned}$$

1 mark for the concept that $\cos\left(\frac{11\pi}{12}\right)$ is < 0

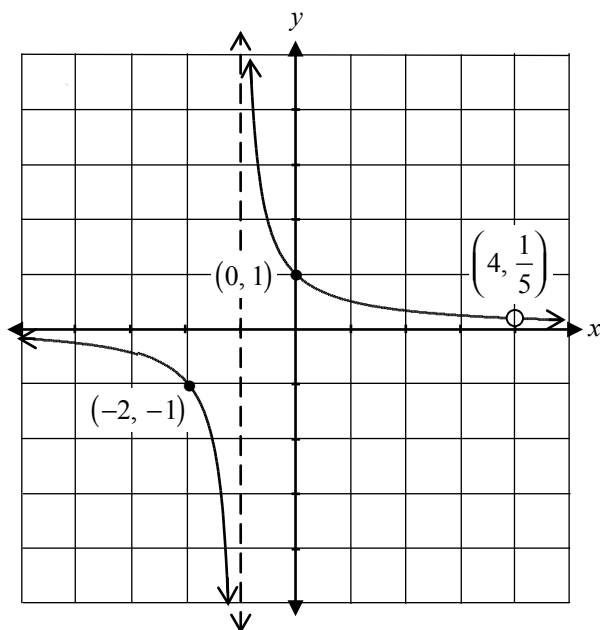
3 marks

Note(s):

- $\frac{-2 - 2\sqrt{3}}{\sqrt{2}}$ is an acceptable way to write the solution
- in Method 1, another possible combination is $\left(\frac{\pi}{6} + \frac{3\pi}{4}\right)$
- deduct ½ mark if answer is not expressed as a single fraction
- in Method 2, the reference angle need not be explicitly stated in order to obtain full marks

Sketch the graph of $f(x) = \frac{x-4}{x^2-3x-4}$.

Solution



1 mark for vertical asymptote at $x = -1$
 $\frac{1}{2}$ mark for graph left of vertical asymptote
 $\frac{1}{2}$ mark for graph right of vertical asymptote
 1 mark for point of discontinuity at $x = 4$

3 marks

$$\begin{aligned} f(x) &= \frac{x-4}{x^2-3x-4} \\ &= \frac{x-4}{(x-4)(x+1)} \\ &= \frac{1}{x+1} \text{ with a point of discontinuity at } x = 4 \end{aligned}$$

point of discontinuity: $f(4) = \frac{1}{5}$

$\therefore$ there is a point of discontinuity at $\left(4, \frac{1}{5}\right)$

$$\begin{aligned} \text{y-intercept: } f(0) &= \frac{0-4}{(0)^2-3(0)-4} \\ &= \frac{-4}{-4} \\ &= 1 \end{aligned}$$

Question 46

R7

Estimate the value of $\log_5 35$.

Justify your answer.

Solution

$$5^2 = 25$$

$$5^3 = 125$$

$\frac{1}{2}$ mark for justification

The value of $\log_5 35$ is more than 2 but less than 2.5.

$\frac{1}{2}$ mark for estimated answer

1 mark

Question 47

R11

If $p(x) = x^5 - 12x + 1$, determine the remainder when $p(x)$ is divided by $(x + 2)$.

Solution

Method 1

$$p(x) = x^5 - 12x + 1$$

$$\begin{aligned} p(-2) &= (-2)^5 - 12(-2) + 1 \\ &= -32 + 24 + 1 \\ &= -7 \end{aligned}$$

1 mark for substitution

1 mark

Method 2

$$-2 \overline{) 1 \ 0 \ 0 \ 0 \ -12 \ 1}$$

1 mark for correct set-up of synthetic division

$$\begin{array}{r} -2 \overline{) 1 \ 0 \ 0 \ 0 \ -12 \ 1} \\ \downarrow -2 \ 4 \ -8 \ 16 \ -8 \\ \hline 1 \ -2 \ 4 \ -8 \ 4 \ -7 \end{array}$$

1 mark

The remainder is -7 .

Describe the effects on the graph of $y = f(x)$ when asked for the graph of $y = f(x - 3) + 5$.

Solution

Shift right 3 and up 5

$\frac{1}{2}$ mark for horizontal translation

$\frac{1}{2}$ mark for vertical translation

or

$(x + 3, y + 5)$

1 mark

Find the exact value of the following expression:

$$\sin\left(\frac{11\pi}{3}\right) \cdot \sec\left(\frac{4\pi}{3}\right) \cdot \tan\left(-\frac{5\pi}{6}\right)$$

Solution

$$= \left(-\frac{\sqrt{3}}{2}\right)(-2)\left(\frac{1}{\sqrt{3}}\right)$$

$$= 1$$

1 mark for $\sin\left(\frac{11\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\sec\left(\frac{4\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\tan\left(-\frac{5\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

3 marks